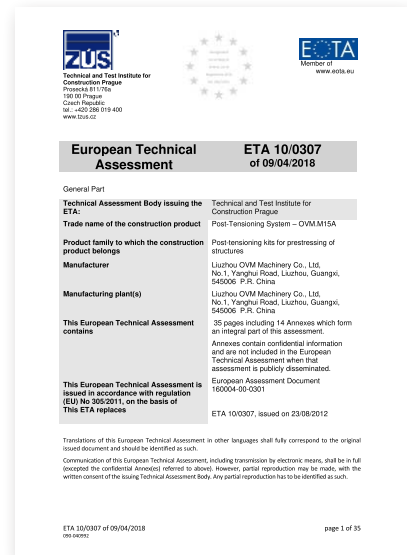
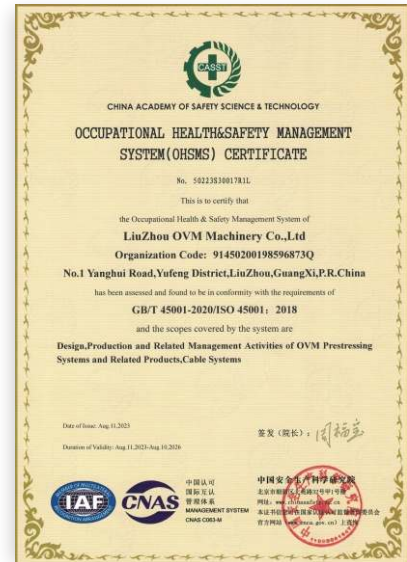


OVM ENGINEERING SOLUTIONS

A large, transparent sphere is the central focus, containing a collage of engineering projects. Inside the sphere, there's a cable-stayed bridge at sunset, a train crossing a bridge, and a cable-stayed bridge over water. The sphere is surrounded by faint, glowing blue lines. At the bottom right, a coiled grey cable leads to a red, square-shaped connector with multiple pins.

Solutions for Bridges, Buildings, Water Conservancy&Hydropower,
Geological Engineering, Containment Structures, Renewable Energy, etc

CERTIFICATES OF OVM



OVM tops the Chinese prestressing industry

With nearly 60 years of experience, OVM is now a leading product supplier and specialist contractor in China in the field of prestressing and other special construction techniques. Enjoying with a high reputation for reliability, professionalism and innovation, OVM's series technology and products have successfully applied on numerous projects, including bridges, highways, high-speed railways, buildings, dams, nuclear power plants, LNG, and have achieved worldwide acknowledgments.

Powerful R&D Capabilities

By implementing it as a basic company strategy, more than 5% of annual our turnover is invested on technical R&D. OVM owns a national technology center and a postdoctoral research workstation and maintains close cooperative relationships with lot of renowned universities and institutions in China. OVM has achieved 1579 technical patents by 2023.

Full range of products

OVM focuses on developing outstanding prestressing technology and building up renowned brand in civil engineering field. As the biggest supplier in China, OVM provides full range of prestressing system including 5 categories, 30 series and more than 420 types of products:

- I. OVM Post-tensioning Systems
- II. OVM Cable Systems
- III. OVM Construction Solutions (Incremental Launching, turning and heavy lifting)
- IV. OVM Bearings & Expansion Joints
- V. OVM Monitoring Systems

Certified management and products

The lasting and reliable OVM products are guaranteed by outstanding management.

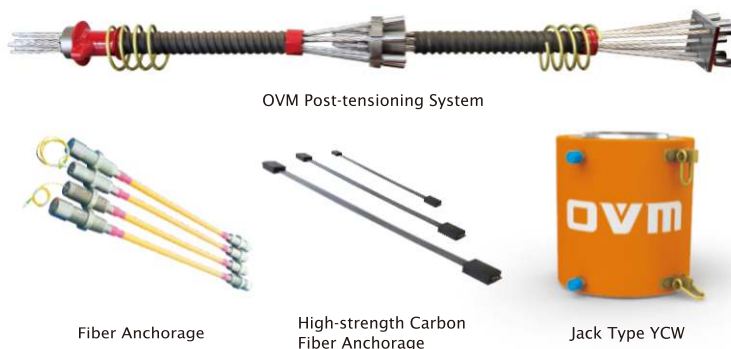
OVM is certified with ISO9001-2015 Quality Management System by BSI and CCC. Each process of production from raw material purchasing to delivery is strictly under the control of management system. OVM products meet the requirements of major standards: AASHTO, ASTM, BS, EN, ETA, FIP, GB, JIS, PTI etc. Meanwhile, OVM works closely with external, independent institutions for testing and improvement of all products.

Yours truly dependable partner

With worldwide network, OVM strives continually to provide high quality services to clients and is seeking partnerships in areas of prestressing design for structural engineering projects, provision of prestressing systems, and contracting of special structures.



OVM POST-TENSIONING SYSTEM



As one of the earliest Chinese post-tensioning systems, OVM has Sufficient experiences in prestressing technology, and the reliability of OVM system is extensively acknowledged in the world.



CENTRAL-WAN CHAI BYPASS, HONG KONG, 2013

Length 4,500m

Erection of precast viaducts, transportation of segments on deck,
supply and installation of post-tensioning



BOUBYAN ISLAND SEAPORT PROJECT (PHASE 1, STAGE 1), KUWAIT

Bridge part includes a 1.42km road bridge and a 4.25km railway bridge across Subiya Straits
2011

Supply of post-tensioning system

- Conforms to world-class standards such as AASHTO, BS, ETA, fib and GB etc.
- Compact anchorage for 1~55 strands
- Adapted for strands with various diameters and strengths

- Easy installation, full series of equipments
- Widely applied to highway, railway, bridge, dam, tunnel, silo and skyscraper etc.



TWRP Tank, Singapore
OVM provides PT materials and stressing equipment.



Mohmand Hydropower Project, Pakistan
OVM supplied ground cable system and stressing equipments



LNG Tank Project, Tianjin, China
2 X 200,000m³ LNG steel storage tank , 8 X 220,000m³ LNG film tank
OVM Supplied cryogenic prestressing system.



Central Business District (CBD), Egypt
The building is going to be the main business district in the new capital of Egypt.
OVM supplied PT System and Equipments, supervision of prestressing works.

EXTERNAL PRESTRESSING SYSTEM



4th bridge of Abidjan, Côte d'Ivoire OVM supplied and installed Internal & External PT system (over 2,000 tonnes in total) as well as the expansion joints.



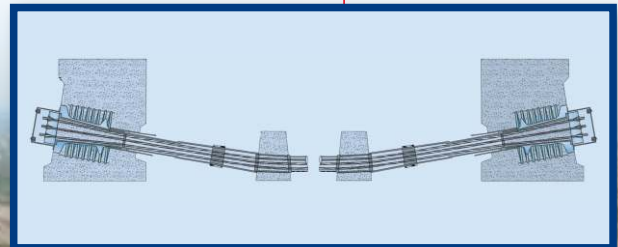
EAST-AFRICA UNITED BRIDGE, TANZANIA & MOZAMBIQUE
Total length 720m, 18 spans, single span 40m
Supply of external prestressing system

SUTONG BRIDGE, CHINA

Length 8,206m, main span 1,088m

Open: 2008

OVM supplied external prestressing and installation services



Guandu Yellow River Bridge (2019)

Main span: 120m

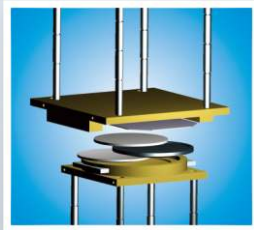
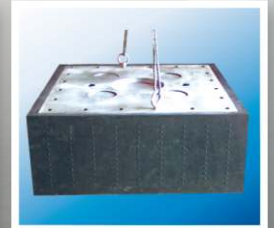
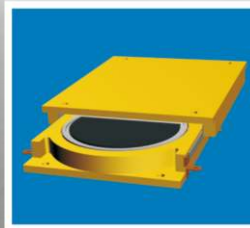
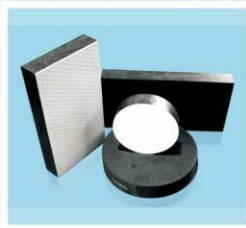
The upper structure of the main bridge features corrugated steel web beams and cantilever cast-in-place beams.

OVM provided the external PT system.

OVM has sufficient experience and numerous job references on external prestressing system.

- New design and application of new materials solve the difficulty of anti-corrosion
- Cable force is measurable and adjustable according to requirement even during bridge operation
- Replaceable system extends the service life of bridge
- Lighter structure, simpler erection and lower cost
- Providing efficient and economical solution for repairing and strengthening of old structures

BEARINGS & EXPANSION JOINTS & DAMPERS



XI'AN XIANYANG INTERNATIONAL AIRPORT HIGHWAY, CHINA
2009
Supply of shock absorbing lead core bearings

OVM provides durable bearings, expansion joints and dampers to various structures:

- Complying with the specifications of JT, AASHTO, BS, JIS etc.;
- Natural rubber and neoprene type of bearings and expansion joints are available;
- Spherical bearings and pot bearings feature high performance;
- Damping bearing with lead core is introduced to projects especially in seismic zone;



WUHAN ER'QI BRIDGE OVER YANGTZE RIVER, CHINA
Length 2,922m, main span 616m
2011
Supply of 408 anti-seismic spherical bearings

Construction works for E60 highway Georgia
OVN supplied of 320 Units LRB bearings and expansion joint.



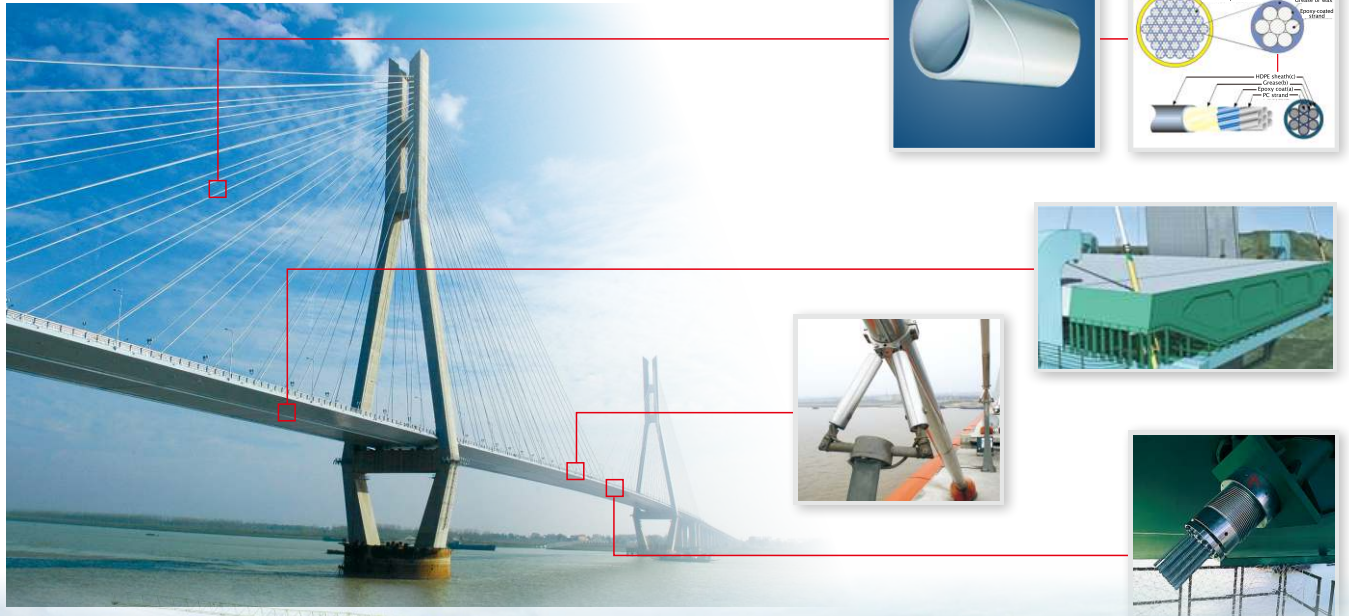
El Ferdan bridge, Egypt
OVN supplied spherical Bearings.



Hong Kong-Zhuhai-Macao Bridge, China, 2016
The longest cross-sea bridge in the world with total length of 55km
OVN supplied high damping bearings, lead rod seismic bearings and $\Phi 75$ large diameter prestressed high strength reinforced bars.

- Shaped steel type and finger type expansion joints provide larger movement and more excellent drive-over comfort;
- Simple, compact, tailor-made Visco Damper is capable of sufficiently absorbing the shock of structure;
- MR Damper provides a continuously controllable, economical solution to wind-rain-induced cable.

CABLE SYSTEMS CABLE-STAYED BRIDGE



ANQING BRIDGE OVER YANGTZE RIVER, CHINA
Length 1,040m, main span 510m 128 stay cables, 1,000 tons
Open: 2004
Supply and installation of stay cable system



Jepak Bridge, Malaysia
Under Construction
OVM supplied stay cables, PT systems, high strength PT bars and expansion joints.

OVM stay cable system solution is based on the applications of multi-strand cable, parallel steel wire cable and strand cable with integral swaging anchorage. OVM has been providing systems and erection services in more than 160 cable-stayed bridges worldwide, all these projects benefit from the excellent performance of OVM solution:



Kinmen Bridge, Taiwan, China
OVM supplied OVM250 Cable System including 432tons of cable



Kigamboni Bridge, Tanzania
Total length: 680meters, Main span : 200meters
Supply of OVM PT System, Bridge Bearing and Expansion Joints,
OVM250 cable system in which the biggest cable is with 160Nos. of
PE sheathed galvanized strands with Dia. 15.7mm.



Rod El Farag Cable Stayed Bridge, Egypt
Scope of OVM: OVM supplied OVM250 stay cable system with
blast and fire protection, and cable force monitoring system,
supervised cable installation and stressing with OVM equipment
and technology.

- Various cable options: bare/galvanized/epoxy-coated strand or wire
- High fatigue performance at 250MPa stress range
- Excellent anchoring property of multi-strand anchorage at the low stress condition
- Higher anti-corrosion performance by applying epoxy-coated strand or wire

- Multi-barrier corrosion protection, satisfying CIP, fib and PTI specifications
- Cable force monitor enables real-time monitoring of stay cables
- Simple structure, easy installation and specialized erection service

CABLE SYSTEMS

CABLE-STAYED BRIDGE



Hwapo bridge, Korea
The main span is 260 meters with two towers and two cable-stayed planes.
PWS Stay Cables were supplied by OVM.



Jiayu Bridge cross Yangtze River, Hubei Province, China
Total length: 4660 M, Main span: 920M, width: 33.5M.
208Nos. of PWS Stay Cables were supplied by OVM.



Yavuz Sultan Selim Bridge, Turkey
Its main span is 1500m and bridge width is 59m which is the widest bridge in the world.
OVM supplied 68 cables and specimen is 109-367 wires.



Ma'anshan Yangtze River bridge for highway&railway, China
Three-tower cable-stayed bridge with main span of 2*1120m.
OVM supply parallel wire stayed cable system with strength of 2100Mpa, and undertake the installation of cables.

Fast-500m Aperture Spherical Telescope, China
OVM supplied 8,895 strand cables to form the cable network of reflector panels, and undertake the installation of cables.



Five-hundred-meter Aperture Spherical Radio Telescope (FAST) Project of National Astronomical Observatories, China

The Biggest Telescope in the world

OVM supplied strand cables to form the cable network of reflector panels and installation services.

Qty. of Cables: 6670 pics

Weight of Cable : 1200MT

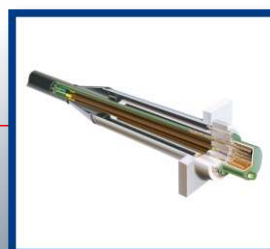
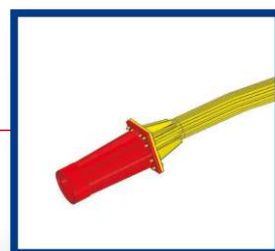
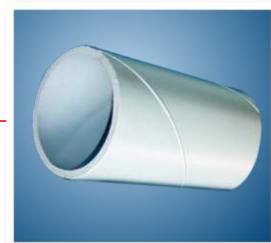
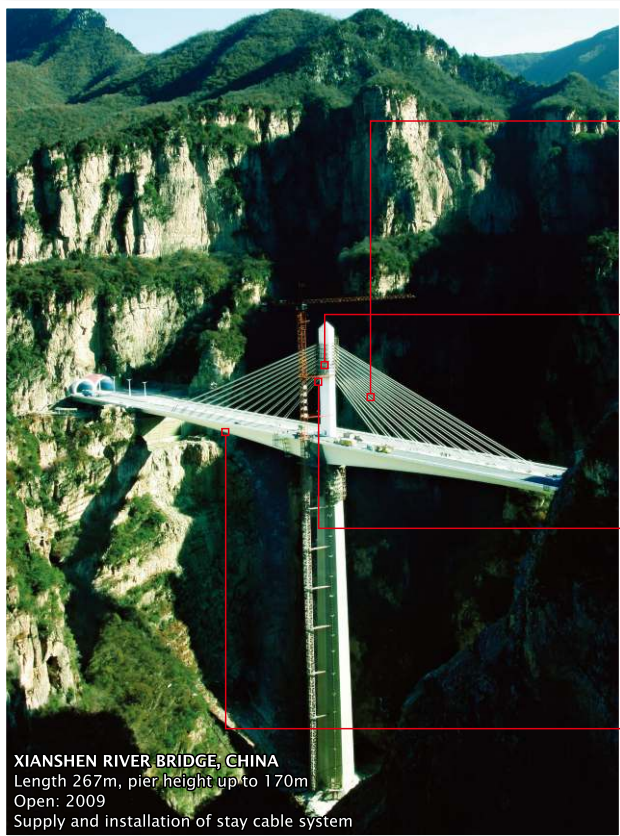
Required Stress Range of Cables: 500Mpa

Required Length Tolerance: +/-2mm

OVM also supplied 100 sets of Intelligent engineering rubber bearings and dynamic health monitoring system for FAST Project.

CABLE SYSTEMS

EXTRADOSSED STAY CABLE BRIDGE

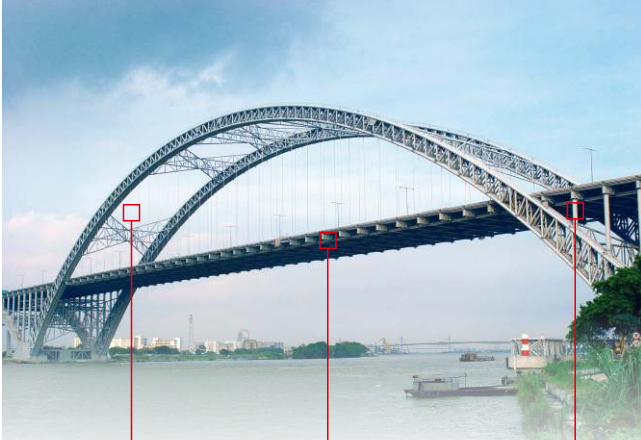


The system is developed especially for extradosed stay cable bridge, which has been applied to more than 500 bridges worldwide.

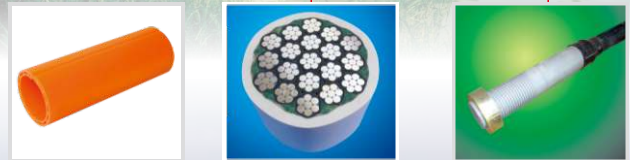
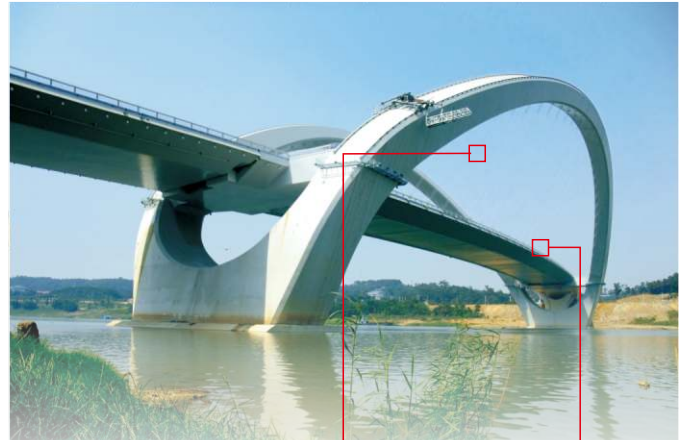
- Outstanding anti-fatigue performance, complying with CIP, fib and PTI standards
- Multi-barrier corrosion protection
- Innovative saddle simplifies the structure and reduces the cost
- Saddle distributes the load and solves the stress concentration
- Easy adjustment and replacement of single strand
- Cable force monitor enables real-time monitoring of cable systems
- Providing complete erection service

CABLE SYSTEMS

ARCH BRIDGE



YAJISHA BRIDGE, CHINA
Length 1,084m, main span 360m
Open: 2000
Supply and installation of hanger and tied cable



NANNING BRIDGE, CHINA
Open: 2009
Total length 1,315m, main span 300m
Supply and installation of hangers and tied cables

3rd Ping Nan Bridge, China
The world's largest span arch bridge with total length of 1,035 meters and main span of 575 meters.
OVM supplied and installed extruded hanger cable as well as smart monitoring system.

OVM has been developed arch bridge solution which is of high durability and adaptability, based on the hanger and tied cable using steel wire or strand.

- New prefabricated PES(FD) hanger improves the property of durability
- Higher anti-corrosion performance by applying epoxy-coated strand or wire
- Newly developed strand cable with integral swaging anchorage has excellent anti-fatigue property
- Cable force monitor enables real-time monitoring of cable systems
- Providing complete erection service
- Customized design for special requirements of structure and safety

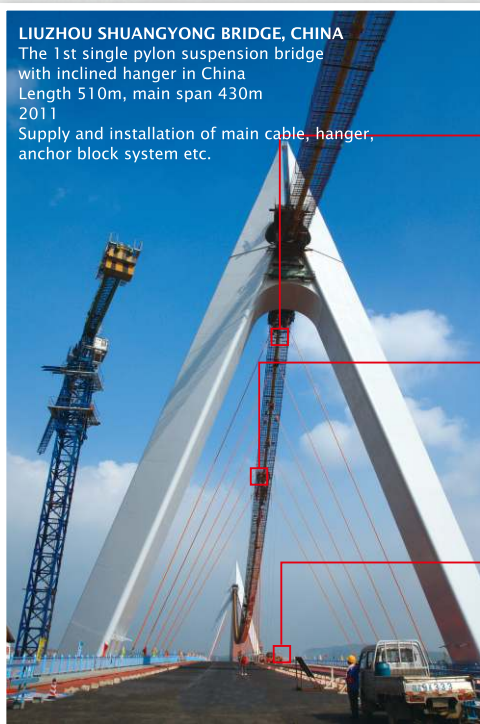
CABLE SYSTEMS SUSPENSION BRIDGE

AIZHAI BRIDGE AT JICHA HIGHWAY, CHINA
Length 1,580m, main span 1,176m
2011

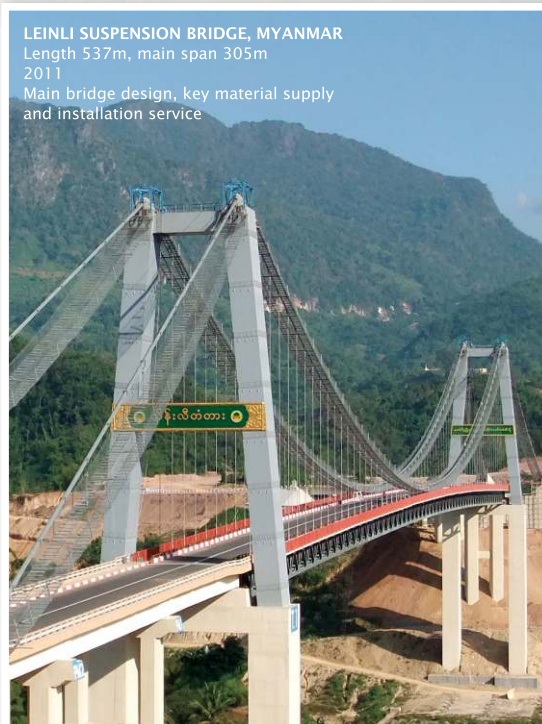
Supply and installation of OVM anchor block system and OVM cable crane



LIUZHOU SHUANGYONG BRIDGE, CHINA
The 1st single pylon suspension bridge with inclined hanger in China
Length 510m, main span 430m
2011
Supply and installation of main cable, hanger, anchor block system etc.



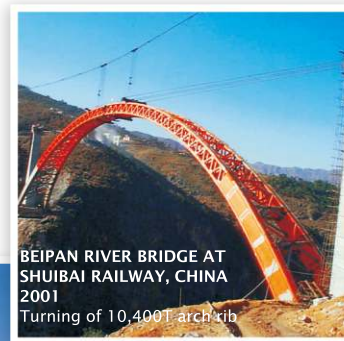
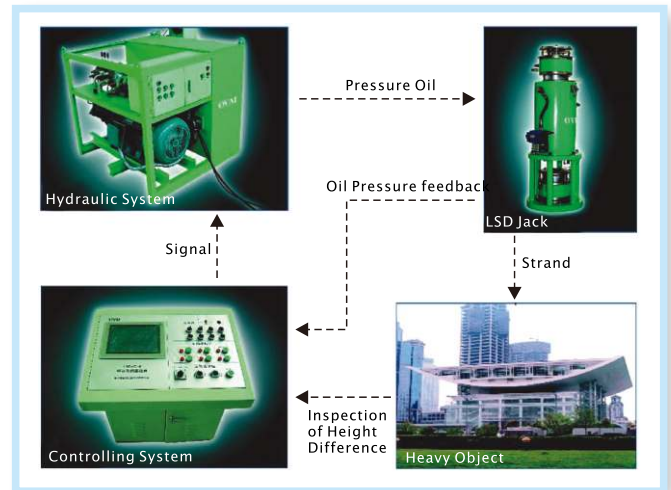
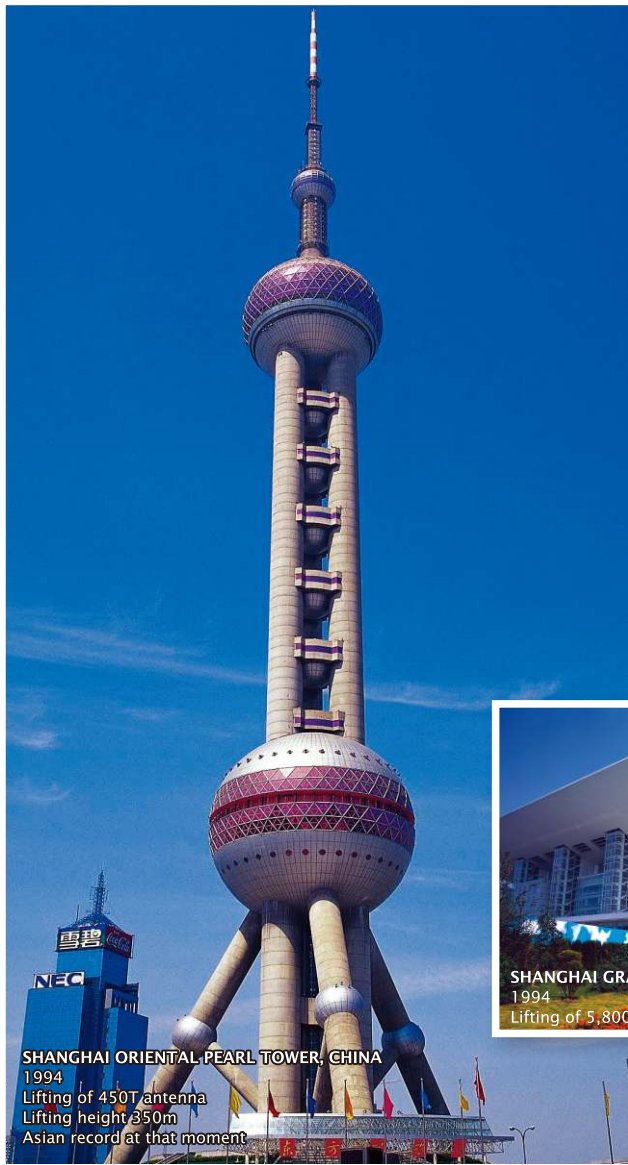
LEINLI SUSPENSION BRIDGE, MYANMAR
Length 537m, main span 305m
2011
Main bridge design, key material supply and installation service



As the main bearing element, cable system for suspension bridge is the lifeline of the bridge. OVM cable system solution for suspension bridge contains complete cable products, which is developed via rigorous scientific research.

- Full range of cable products for suspension bridge ensuring perfect match
- Advanced replaceable anchor block system with excellent anti-fatigue and anchoring performance
- PPWS main cable and hanger with outstanding anti-corrosion property
- Offering optimization to cable saddle and band etc.
- Providing 3 major equipments for construction of suspension bridge: cable crane, wrapping machine and main cable compressor, capable of contracting the special construction process
- Cable force monitor enables real-time monitoring of cable systems if necessary

HYDRAULIC SYNC LIFTING TECHNOLOGY



OVM hydraulic sync lifting technology features innovative design, special construction method, advanced automatic control and high reliability, which has been involved in several great projects. It is an excellent solution to move the extra large, ultrahigh, superheavy structures in horizontal and vertical direction, which is also widely applied to bridge turning.

- Easy and rapid installation
- No limit to operating surface and mass of the structure being lifted (incrementally launched or turned)
- Highly automatic, the deviation of pose, velocity, lifting or traction force of the structure could be controlled accurately
- Simple operation, visualized parameters, high reliability
- Decrease working amount at height, shorten construction period, improve erection quality, lower the cost

MONITORING



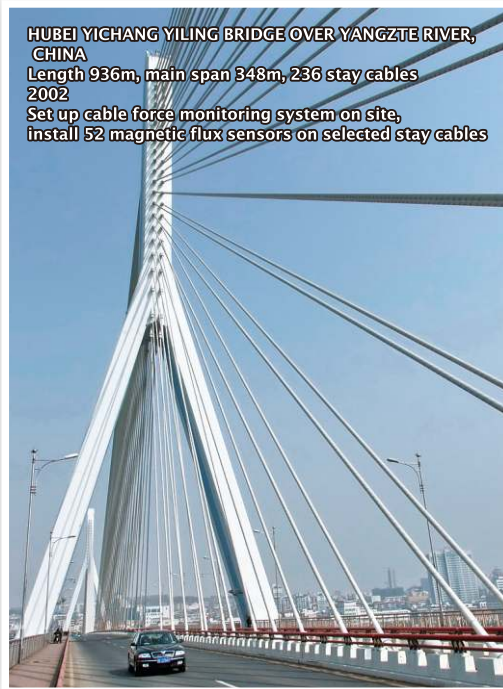
YIBIN BRIDGE OVER YANGZTE RIVER, CHINA
Length 850m, main span 460m
2008
Set up 2 sets of GPRS online cable force monitoring systems, install 80 magnetic flux sensors on selected stay cables



HONG KONG STONECUTTERS BRIDGE, CHINA
2009
Total length 1,596m, main span 1,018m
Supply and install magnetic flux sensors to monitor the external cable force



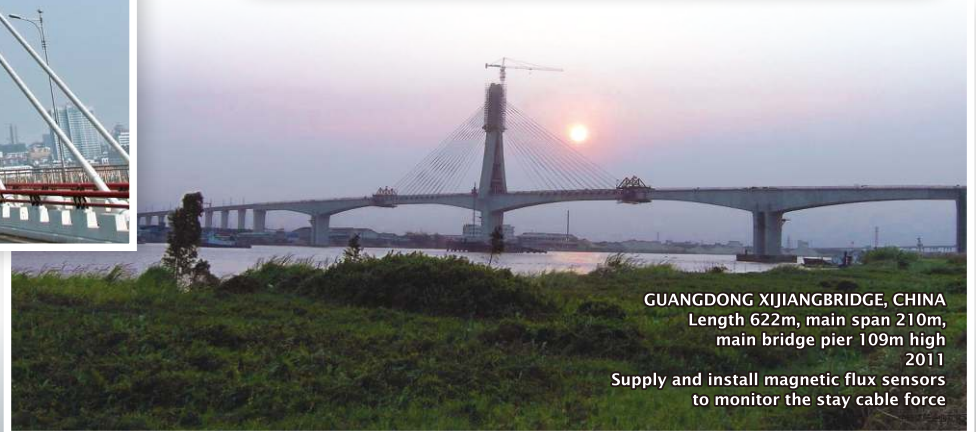
JIANGXI JIANYI BRIDGE, CHINA
Length 440m, main span 165m
2008
Set up bridge operation management system, monitoring force of stay cable, stress of pylon and girder, and structural data such as displacement, deformation, vibration, temperature and humidity etc.



HUBEI YICHANG YILING BRIDGE OVER YANGTZE RIVER, CHINA
Length 936m, main span 348m, 236 stay cables
2002
Set up cable force monitoring system on site,
install 52 magnetic flux sensors on selected stay cables

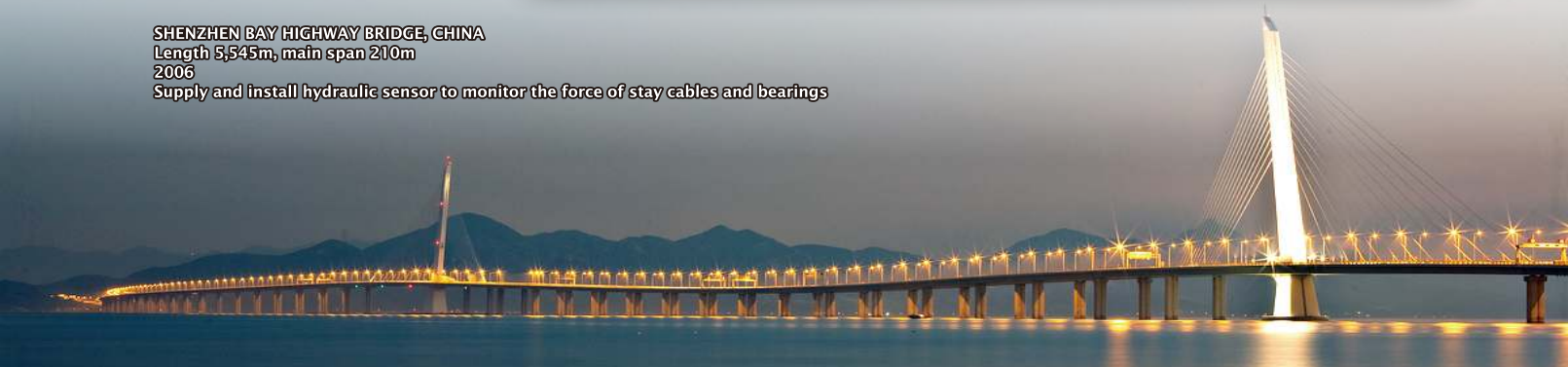


ARCH BRIDGES ON BEIJING-SHANGHAI & GUANGZHOU-SHENZHEN-HONG KONG HIGH-SPEED RAILWAY
Supply and install magnetic flux sensors to monitor hanger force of more than 10 arch bridges on Beijing-Shanghai & Guangzhou-Shenzhen-Hong Kong high-speed railway.

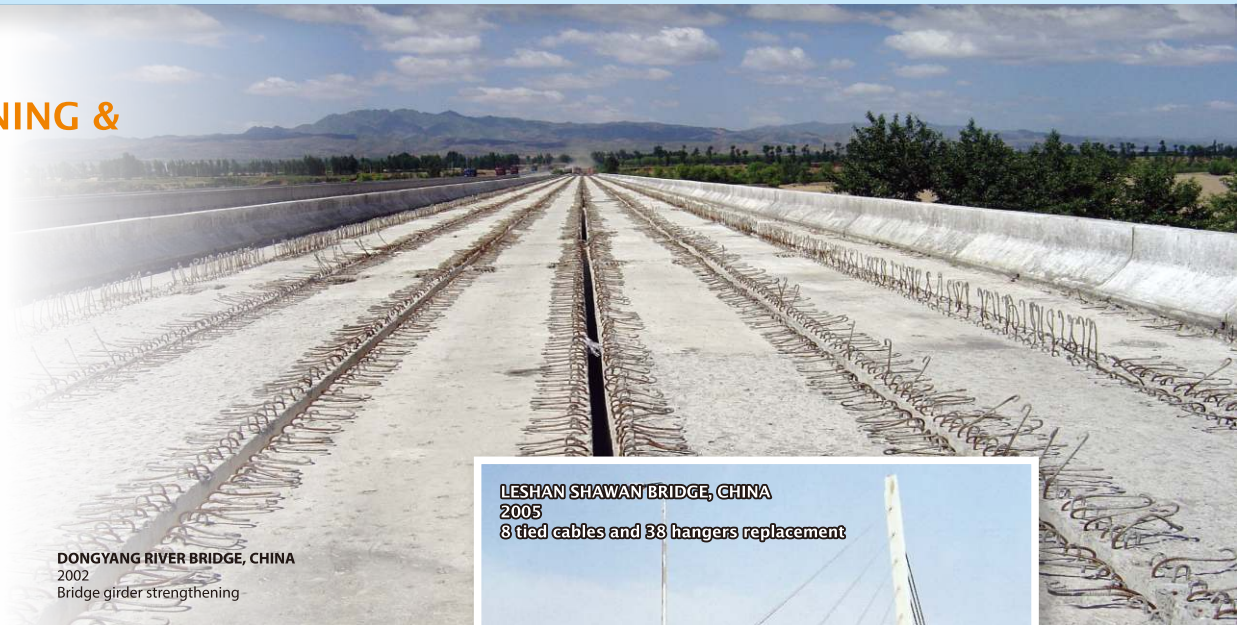


GUANGDONG XIJIANG BRIDGE, CHINA
Length 622m, main span 210m,
main bridge pier 109m high
2011
Supply and install magnetic flux sensors
to monitor the stay cable force

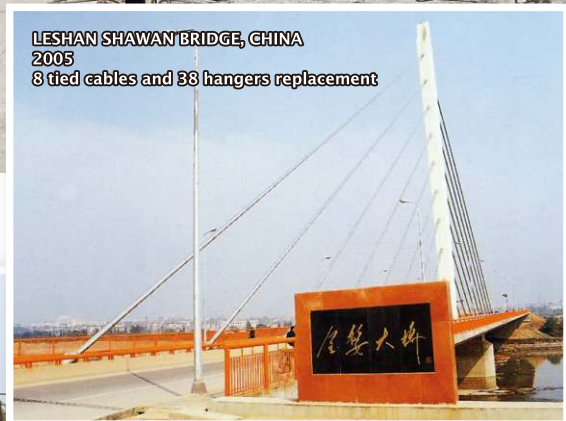
SHENZHEN BAY HIGHWAY BRIDGE, CHINA
Length 5,545m, main span 210m
2006
Supply and install hydraulic sensor to monitor the force of stay cables and bearings



STRENGTHENING & REPAIRING



DONGYANG RIVER BRIDGE, CHINA
2002
Bridge girder strengthening



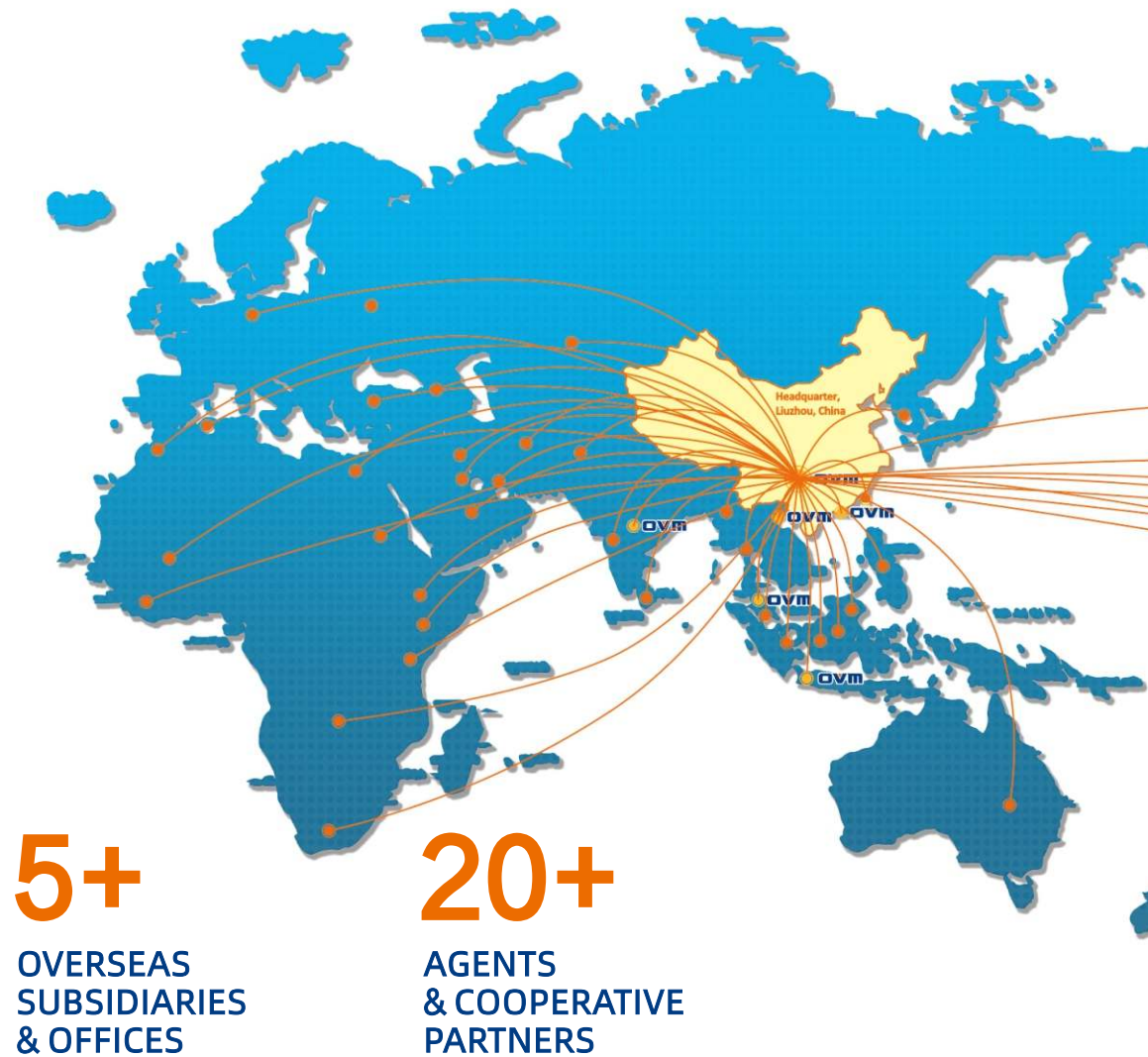
LESHAN SHAWAN BRIDGE, CHINA
2005
8 tied cables and 38 hangers replacement



JINHUA JINWU BRIDGE, CHINA
2005
Stay cable replacement



OVM Global Network





1500+

CONSTRUCTION
PROJECTS IN OVER
60 COUNTRIES

**REALIZE YOUR IDEAS,
SUPPORT YOUR SOLUTIONS.**



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